

Vinh Ly

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Summary

Software Engineer with **18 years** delivering end-to-end high-throughput services for video, e-commerce, and cloud platforms. Owned the live-streaming pipeline serving 250 M daily video impressions, the real-time auction engine handling \$1.2 B transaction volume, and the distributed ad-serving stack powering billions of requests. Core stacks include **C++**, **Go**, **Python**, **Java**, and **Kubernetes**. Cut processing latency by **32 %** while scaling to 5× traffic through kernel-level optimizations. Partnered with product, data, and security teams to ship features on a fortnightly cadence. Next focus: scaling compute and storage primitives for next-generation cloud workloads.

Core Skills

Programming Languages: C++, Go, Python, Java, Rust, TypeScript, JavaScript, HTML, CSS, SQL

Systems & Orchestration: Kubernetes, Linux, gRPC, Protobuf, REST

Performance Engineering: Profiling, Tracing, Benchmarking, Concurrency, Cache Design

Cloud & DevOps: Google Cloud Platform, CI/CD, Git, Terraform, Distributed Systems

Work Experience

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007

Staff Software Engineer, YouTube (02/2025 - Present)

- Re-architected the live-chat service in **C++** using lock-free queues, enabling 3× concurrent users and reducing tail latency by **28 %**.
- Integrated **Kubernetes** autoscaling with custom metrics, increasing request throughput by **45 %** during peak events while maintaining sub-second response times.
- Led a cross-functional effort with product and security teams to harden video ingest pipelines, cutting security incidents by **70 %** over twelve months.
- Implemented **gRPC**-based micro-services for thumbnail generation, decreasing processing time from 120 ms to 68 ms, a **43 %** improvement.
- Mentored a squad of 8 engineers on kernel-level profiling techniques, shortening debugging cycles by an average of **2 days** per incident.

Staff Software Engineer, Whatnot (05/2024 - 02/2025)

- Built a real-time auction engine in **Go** that handled \$1.2 B in transaction volume, improving order matching latency by **31 %**.
- Designed a **Kubernetes**-native deployment model for live-streaming services, achieving a **4×** reduction in infrastructure cost per stream.
- Collaborated with data science partners to embed predictive pricing models, raising average auction final price by **12 %**.
- Refactored legacy **Python** batch jobs into asynchronous pipelines, cutting nightly processing time from 6 hours to 2 hours (**66 %** reduction).
- Established end-to-end observability using **gRPC** tracing, decreasing mean time to detection from 45 minutes to 9 minutes.

Staff Software Engineer/Manager, Google (03/2015 - 04/2024)

- Directed a team of 12 engineers to launch a distributed ad-serving platform in **Java**, delivering 2.3 B requests daily with **99.97 %** availability.
- Optimized cache eviction policies in **C++**, reducing cache miss rate by **22 %** and saving \$3 M in compute spend.
- Introduced **Kubernetes**-based canary deployments, shortening release cycles from two weeks to three days while maintaining service stability.
- Partnered with product managers to define SLA targets, achieving a **15 %** improvement in page load speed across flagship services.
- Implemented a cross-team **CI/CD** pipeline using Git, cutting build time by **40 %** and enabling multiple daily releases.

Software Engineer, Google (02/2011 - 02/2015)

- Created a high-throughput logging pipeline in **Python**, ingesting 500 TB of data per month with **99.9 %** reliability.
- Enhanced search indexing service in **C++**, boosting query throughput by **27 %** while keeping latency under 50 ms.
- Worked with security engineers to integrate **TLS** encryption across internal APIs, eliminating 18 % of reported data-exposure incidents.
- Automated deployment of micro-services using **Docker** and **Kubernetes**, reducing manual configuration errors by **85 %**.
- Collaborated with data analysts to build real-time dashboards, improving operational visibility and cutting incident response time by **30 %**.

Software Design Engineer II, Microsoft (10/2007 - 12/2011)

- Led the redesign of a cloud storage gateway in **C++**, increasing throughput by **38 %** and supporting 1.5 M concurrent connections.
- Implemented **REST** APIs for file synchronization, reducing client-side latency by **24 %**.
- Co-authored a **gRPC** communication layer that cut inter-service latency by **19 %**.
- Optimized memory allocation patterns, decreasing peak RAM usage by **22 %** and extending hardware lifecycle.
- Partnered with QA teams to establish automated regression suites, improving test coverage to **92 %**.

Software Design Engineer Intern, Microsoft (06/2006 - 08/2006)

- Developed a prototype **Python** script to automate log aggregation, cutting manual effort by **80 %**.
- Assisted in performance profiling of a legacy **C++** module, identifying bottlenecks that reduced execution time by **15 %**.
- Collaborated with senior engineers to refactor a data-parsing routine, improving code readability and lowering defect density.
- Presented findings to the product team, influencing the roadmap for the next release cycle.